

DEFICITS IN STRATEGY APPLICATION FOLLOWING FRONTAL LOBE DAMAGE IN MAN

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SUMMARY

A quantitative investigation of the ability to carry out a variety of cognitive tasks was performed in 3 patients who had sustained traumatic injuries which involved prefrontal structures. All 3 had severe difficulties in 2 tests which required them to carry out a number of fairly simple but open-ended tasks over a 15–30 min period. They typically spent too long on individual tasks. All patients scored well on tests of perception, language and intelligence and 2 performed well on a variety of other tests of frontal lobe function. Explanations for their difficulty on the multiple subgoal tasks in terms of memory or motivational problems could be excluded. It is argued that the problem arose from an inability to reactivate after a delay previously-generated intentions when they are not directly signalled by the stimulus situation.

INTRODUCTION

It has been known for many years that frontal lobe lesions can produce a gross effect in the performance of everyday life activities other than the most routine, even though neuropsychological tests suggest that the cognitive changes that have occurred are at most minor (e.g., Brickner, 1936; Ackerly and Benton, 1947). For instance, Eslinger and Damasio (1985) reported the case of an accountant who 6 yrs after an operation for the removal of a large bilateral orbitofrontal meningioma had an IQ of over 130 and performed well on a wide variety of neuropsychological tests, including some held to be sensitive to frontal lobe damage. Despite this excellent performance on quantitative tests, however, his ability to organize his life was grossly impaired. He was dismissed from a series of jobs even though his basic skills, manner and temper were appropriate. He went bankrupt and was involved in two divorces in 2 yrs. Relatively simple matters would take hours; thus to go out to dinner required that he consider the seating plan, menu, atmosphere and management of each restaurant and he might even drive to see how busy each of them was, but was still unable to come to a decision.

Of course, patients with relatively severe frontal lesions can have deficits on a range of quantitative neuropsychological tests (*see* Stuss and Benson, 1986, for review). It is possible that all patients who have debilitating everyday life difficulties would also show severe quantitative deficits on standard neuropsychological tests, but that the relevant tests were not given to the patient. However, a more probable explanation is that certain of the implicit approaches adopted in the design of neuropsychological tests makes the patient's impairment less likely to manifest themselves in the test situation. Thus in neuropsychological tests the patient typically has a single explicit problem to tackle at any one time, the trials tend to be very short (1 min or so or even less), task initiation

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is strongly prompted by the examiner and what constitutes successful trial completion is clearly characterized. Rarely are patients required to organize or plan their behaviour over longer time periods, or to set priorities in the face of two or more competing tasks. Yet it is these sorts of 'executive' abilities which are a large component of many everyday activities.

For instance, consider a situation which gave problems for a patient who had had a right frontal glioma removed and who was Wilder Penfield's sister. In Penfield and Evans (1935) he writes: 'She had planned to get a simple supper for one guest (Penfield) and four members of her family. She looked forward to it with pleasure and had the whole day for preparation. When the appointed hour arrived she was in the kitchen, the food was all there, one or two things were on the stove, but the salad was not ready, the meat had not been started and she was distressed and confused by her long continued effort alone.' If in addition to the actual preparation of the meal one also has to decide the menu, then an everyday task such as this will require certain capacities, none of which are clearly captured in the standard neuropsychological test. Planning will have to be carried out and preparatory purchases made. Different considerations—costs, tastes, available time, etc.—may have to be weighed against each other. Many minor decisions will need to be made and typically they are undertaken in parallel with other activities. There is no clearly correct solution and many different activities may have to be scheduled over the time available.

Before this second explanation can even be seriously considered, however, it is necessary to develop quantifiable analogues of the open-ended multiple subgoal situations where this subset of frontal patients would theoretically have problems. Only then can this possibility be realistically considered. In this paper we describe 3 patients with severe problems in the organization of everyday life activities arising from head injuries primarily affecting the frontal lobes. Their performance on a large set of standard neuropsychological tests ranged from generally good with patchiness on some 'frontal' tests in the worst of the 3 patients, through to almost consistently bright average to superior performance in the best of the 3. However, all 3 patients performed especially poorly on two tests designed to assess performance in more open-ended multiple subgoal situations. Various alternative explanations for their poor performance are then assessed.

CASE HISTORIES

Case 1

A.P., a right-handed man was involved in a road traffic accident at the age of 23 yrs, approximately 4.5 yrs before the present investigation. He sustained a serious open head injury involving an anterior fossa fracture, requiring multiple operations. His condition gradually improved, but he was not able to return successfully to his job. A CT scan that year showed evidence of extensive bifrontal damage. A year later he was admitted as a day patient to St Andrew's Hospital for rehabilitation. He was well-motivated and keen but could not carry out even the simplest activity because of an inability to keep his mind on the task in hand. For example, on one occasion he was discovered on the local golf course having originally stepped outside the therapy room to fetch some coffee. At these times he maintained that he 'completely forgot' whatever it was that he was supposed to be doing. He was unable to shop for himself because he would buy one item at a time returning to his car after his every purchase.

Rehabilitation as a day patient had limited success, and after 3 months A.P. was transferred to another rehabilitation centre as an inpatient. There he was treated with behavioural methods described by Wood and Burgess (1988) with a slow but progressive improvement in his ability to organize his activities of

daily living. He remained in rehabilitation for approximately 1 yr and then returned home to live with his parents.

In response to a clinical interview, he said that before the accident his room was immaculate ('5/5 tidiness') but it is now untidy with 'hotchpotch piles of magazines' on the floor. He had had an efficient filing system but had abandoned it. Shopping, cleaning and laundry are done for him. His social life is very rarely organized in advance. When questioned he said he had nothing planned for the coming weekend, was unable to give any example of consciously organizing an activity beforehand.

Clinical neuropsychological investigations. A.P. was tested at St Andrew's Hospital and then 2 yrs later, at the National Hospital on a large range of neuropsychological tests. On almost all he performed very satisfactorily. At St Andrew's Hospital he obtained a Verbal IQ of 121 and a Performance IQ of 120, which corresponds well with the estimate of 124 for his premorbid IQ as measured by the NART. There was one exception—the Digit Span subtest where he scored only in the Dull Normal range; however, on a repeat testing 2 yrs later he scored in the Superior range on the subtest. His performance on other subtests was slightly better, which is probably a practice effect (*see* Table 1). His spontaneous speech was not aphasic and on all perceptual, language, arithmetic and memory tests carried out he performed satisfactorily (*see* Tables 1, 2). The one memory test where his performance was below average was the Petrides and Milner (1982) self-ordered pointing task which is most appropriately considered a frontal test. On the picture version he made 7 errors by comparison with a mean of 3 errors for their control subjects who were of comparable age. (However, he was within 1 SD of the mean of a posterior lesion control group (*see* Table 3).) On all of a sizeable group of other tests thought sensitive to frontal lesions he performed very satisfactorily (*see* Table 3).

TABLE 1. PERFORMANCE ON BASELINE TASKS
Wechsler Adult Intelligence Scale (WAIS) Subtest Age-Scaled Scores

	Cases				Cases		
	1	2	3		1	2	3
Verbal							
Arithmetic	13	14	15	Picture Completion	14	12	13
Similarities	15	14	15	Block Design	15	12	13
Digit Span	15	14	15	Picture Arrangement	13	11	11
Vocabulary	16	15	17				
Verbal IQ	128	126	135	Performance IQ	129	112	114
	Cases						
		1	2	3			
WAIS Full Scale IQ		130	121	127	(Wechsler, 1955)		
NART FSIQ equivalent		124	119	127	(Nelson and O'Connell, 1978)		
Language (Scaled Score)		A.P.	D.N.	F.S.			
Graded Naming Test		14	14	14	(McKenna and Warrington, 1983)		
Perception (percentiles)		A.P.	D.N.	F.S.			
Cube Analysis		50	50	50	(Warrington and James, 1988)		
Dot Centre		50	50	50	(Warrington and James, 1988)		
Unconventional Views		50	25	10	(Warrington and Taylor, 1973)		
Conventional Views		50	50	50	(Warrington and Taylor, 1973)		

Case 2

D.N. a right-handed professional man, had sustained a serious head injury in a road traffic accident at the age of 26 yrs, involving a right frontal depressed skull fracture and an intracerebral haematoma treated surgically. He still has severe left hemiparesis. He says that consciousness was reduced or altered for 3 months. A CT scan carried out at age 48 yrs showed an extensive low attenuation area in the right frontal lobe and marked local atrophy of the right medial insular cortex. Both lateral ventricles were enlarged, the right more than the left. There were additional mild changes in the left frontal lobe. On leaving hospital after a stay of 6 months he returned to his previous employment but was unable to carry it out. He failed a higher degree but obtained a teacher's certificate. For the next 5 yrs he held a succession of jobs from most of which he was dismissed.

His responses in a clinical interview and his wife's testimony were in agreement over his domestic behaviour. He is untidy. He only bathes if going somewhere important. Shaving, changing his clothes or undergarments, washing his hair and having his hair cut are only carried out when his wife tells him. He hardly ever spontaneously tackles any domestic chores such as laundry, cleaning, cooking, making repairs or paying bills. If his wife is out he normally leaves the preparation of a meal to his 10-yr-old son. When he shops he never makes out a list himself and also usually comes home without all the items on the list his wife prepares. When she gives him a task she has to specify exactly what is required and even so he might carry out some parts only and then starts reading a newspaper. In addition, his wife organizes all trips, outings and social contacts with relatives. His wife said that he was occasionally irresponsible over money; for instance, even though they were in financial difficulties as he was out of work, he would buy gadgets they did not really need, including a sophisticated music system, costing £500.

Clinical neuropsychological tests. D.N. was tested in the National Hospital on a wide range of neuropsychological tests. On the WAIS, he obtained a Verbal IQ of 126 and a Performance IQ of 112, which may well be a little below his premorbid level but in fact corresponds well with the results of the National Adult Reading Test (Nelson, 1983) of 119. For all Verbal subtests he was in the superior range and in the Performance subtests in the average or bright average ranges (see Table 1). On memory tests he performed well with verbal material but poorly with visual material (see Table 2). On tests held to be sensitive to frontal lobe lesions his performance was generally satisfactory, but it was rather poor on the picture version of the Self-Ordered Memory test (see Table 3).

In summary, D.N. performed well on a wide range of perceptual language and frontal lobe tests. However, while his performance on verbal memory tests was well within the normal range, it was impaired on visual memory tasks.

Case 3

F.S., a 55-yr-old right-handed woman, employed in an undemanding post, as a 'media resources officer', had earlier sustained two separate head injuries. Thirty years before she had been thrown from a horse, had fractured her skull and had been left with a posttraumatic amnesia of unknown length. She was also aphasic at the time. The accident had left her with slight left-sided facial paralysis. Two years before being tested she had been knocked off her bicycle by a car and hit her head on the road; she did not lose consciousness but as a result has had a permanent loss of her sense of smell.

A CT scan carried out 2 yrs after the second accident showed an extensive lesion to the left frontal lobe with atrophy causing enlargement of the frontal horn of the lateral ventricle. There was also some atrophy in the left temporal lobe.

For the past 25 yrs she has worked in the same position. She lives by herself in a single room. Her responses in a clinical interview show that she undertakes virtually no inessential or novel activities. She is very untidy, never putting things away. She seldom goes out in the evening, and virtually never travels away from her home town. Others always make arrangements when any joint activity is to be carried out. She is said by her sister never to organize anything. She shops every day buying only a few things on any occasion and never visits supermarkets. She had no activity planned for the following weekend and could give no example where anyone had relied on her to do anything. Her sister confirmed that these behaviours were characteristic.

Clinical neuropsychological tests. F.S. was tested at the National Hospital on a large range of neuropsychological tests. On the WAIS she obtained a Verbal IQ of 135 and a Performance IQ of 114 comparable with the estimate of the premorbid IQ of 127 derived from the New Adult Reading Test. Verbal

TABLE 2. PERFORMANCE ON TESTS OF MEMORY (SCALED SCORES)

Test	Cases		
	1	2	3
Auditory Verbal Learning ¹	12 ²	8	9
Complex Figure Recall ¹			
Immediate	NT*	NT	6
Delayed	17 ³	3 ³	6
Digit Span ⁴	15	15	15
Paired Associates (Sum) ⁵	11	NT	7
Recognition Memory ⁶			
Words	10	10	15
Faces	14	4	11
Story Recall ¹			
Immediate	13 ⁵	11	11
Delayed	14 ⁵	10	12
Visual Reproduction ⁵	13	14	NT

* NT = not tested. ¹ Coughlan and Hollows (1985). ² Score on Worst Trial of Rey Auditory Verbal Learning Test (Rey, 1964). ³ Osterrieth (1944–1945). ⁴ Wechsler (1955). ⁵ Wechsler (1945). ⁶ Warrington (1984).

TABLE 3. PERFORMANCE ON 'FRONTAL LOBE' TASKS

	Cases			Comment
	1	2	3	
Alternation task ¹ (trial of last error)	90%ile	75%ile	5%ile	Compared with nonlesion patient/controls ²
Bilateral hand movements	Good	Good	Good	
Cognitive Estimates ³ (error score)	2	4	5	Mean of controls = 3.0, SD 2.6 ²
Money's Road-Map Test ⁴ (errors)	0	3	5	Mean of left frontal group = 11.9; normals = 1.7
Personal Orientation Test ⁵ (errors)	0	4	14*	Mean of right posteriors = 4.1, SD = 3.5 Mean of left posteriors = 2.4, SD = 2.7 ⁶
Proverb Interpretation	Good	Good	Good	
Self-Ordered Memory ⁷ Representational Pictures (errors)	7	10	9	Cases 2 and 3 > 1 SD worse than a lesion control group ⁸ (mean = 5.9, SD 2.8)
Stroop ⁹ (time)	20.1	17.3	21.0	Control: mean = 23.3, SD = 11 ²
Tower of London (score) ¹⁰	33	23	24	Controls: mean = 24.9, SD = 4.5 ²
Trail Making: Letters and Numbers ¹¹ (completion time in s)	15.7	25.6	49.0*	Controls: mean = 29.6 s, SD = 4.5 ²
Verbal Fluency: Letters FAS, each 60 s (total retrieved)	70	34	33	Controls mean = 34.8, SD = 14.3 (n = 30) ¹²
Modified Wisconsin ¹³ (categories achieved)	6	6	6	Controls mean = 5, SD 1.6
Total errors	3	3	1	Controls mean = 9.2, SD 8.5 (n = 46; mean age (yrs) 45, SD 14)

* More than 2 SD worse than controls. ¹ Chorover and Cole (1966). ² Nonlesion patient controls (n = 24, mean FSIQ = 114.9, SD 12.7): Shallice, Warrington, Watson and Lewis (unpublished study). ³ Shallice and Evans (1978). ⁴ Butters *et al.* (1972). ⁵ Semmes *et al.* (1963). ⁶ 10 right posterior lesion patients (mean FSIQ 108.2, SD 9.8, mean age 45.7 yrs, SD 14.0) and 10 left posterior lesion patients (mean FSIQ 108.1, SD 12.2, mean age 41.8 yrs, SD 14.0). ⁷ Petrides and Milner (1982). ⁸ 8 right posterior patients (mean FSIQ = 107.1, SD 8.9, mean age 45.5 yrs, SD 14.8. Mean Scaled Score FC Faces = 7.25, SD 4.8 (dull average)). ⁹ Perret (1974). ¹⁰ Shallice and McCarthy (*see* Shallice, 1982). ¹¹ Reitan (1958). ¹² Miller (1984). Note, however, that if Miller's formula for predicting fluency from WAIS verbal subtests is used, Cases 2 and 3 both fall below the expected range. ¹³ Nelson (1976).

subtests were all carried out at the Superior level. Visual subtests were carried out at the Bright Normal level except for Picture Arrangement which was at the average level. She had no aphasic difficulties and performed in the normal range on language and perceptual tests (*see* Table 1).

Her performance on memory tests was mixed (*see* Table 2). However her pattern of spared/impaired performance, with difficulty on learning span +1, which she was unable to do in 12 trials for both digits and spatial positions (Corsi span (*see* Milner, 1971)); also on the Rey Figure, in the context of spared Recognition Memory, Logical Memory and free recall suggests that she has a memory problem secondary to frontal difficulties (Signoret and Lhermitte, 1976).

On tests thought sensitive to frontal lobe damage her performance was variable (*see* Table 3). On a number it was satisfactory, but on 4 tests it was impaired. Thus both memory and cognitive tasks showed evidence of frontal impairment.

Summary of patients

Overall the 3 patients perform well and close to their premorbid level on a wide range of perceptual, language and general cognitive tests. Case 1 has no problem with memory tasks but Case 2 has a primary visual memory impairment and Case 3 has difficulties on certain tasks of a 'frontal memory' type. Two of the patients perform very well on a wide range of tasks thought sensitive to frontal lobe lesions but 1 shows impairments on some frontal tasks. All show severe organizational difficulties in everyday life.

NEUROPSYCHOLOGICAL TESTS

Multiple subgoal scheduling tests

Two tests were developed to measure the ability of a subject to schedule the performance of a number of simple tasks when the demands made are rather open-ended and there are a number of simple rules which apply across all tasks.

The Six Element Test (SE). The purpose of this test, which is undertaken in a standard hospital office, is to assess the ability of the patients to carry out 6 open-ended tasks in a fixed period of time (15 min) in a way that maximizes their overall score. The 6 tasks divide into 2 sets of 3—dictating a route, carrying out arithmetic problems and writing down the names of approximately 100 pictures of objects. Each set of arithmetic problems contains 35 problems generally of increasing difficulty. Thus in set A,

$$\text{no. 8 is } \frac{300}{6} \times 4, \text{ no. 18 is } 201 + 187 - 86 \text{ and no. 28 is } \frac{210}{7} \times 8 + (6 \times 4).$$

The 2 sets are equivalent in difficulty.

The subject is presented with the instructions written on a card (*see* Appendix 1). The examiner (E) first reads through the complete test instructions with the subject (S), explaining which stacks of cards are piles A and B and which arithmetic set is I and II and so forth. Having read through the task descriptions the E says 'But before you go on, there are a number of important rules you *must* follow which govern the way you are allowed to do these tasks, and may influence the way in which you decide to organize the way you do the test'. The 5 rules are read through with the S and explained as fully as necessary. The instruction sheet is then turned over and the S is 'tested' on his recall of the rules. Any errors are corrected. If she/he has forgotten any, the entire rules are read through and then the S is tested again. This procedure is repeated until all the rules are recited correctly. It is explained to the S that the instructions will be left in front of them at all times to consult at will.

Two subsidiary tests were also undertaken. First, the subjects, whose watches had been removed, had to carry out the Digit Symbol subtest of the WAIS, stopping themselves after 1 min 15 s. A stop watch was visible throughout. Secondly, Rey's (1964) Tangled Lines test had to be carried out with the subjects instructed to stop as close as possible to 2 min 30 s after starting. This time the stop watch was covered by a small cardboard box. The box could be moved whenever the subjects wanted to know the time but had to be immediately replaced. With both these tests, were the Ss not to stop themselves at the appropriate time the E made no indication that they were to stop, and they were allowed to continue.

Controls. 10 control subjects were tested. They were matched to the patients on age (mean 41, range 25–62 yrs), educational background, and NART IQ (mean 120.2, range 113–127 yrs).

TABLE 4. PERFORMANCE ON THE SIX ELEMENT TASK

<i>Case</i>	<i>No. of subtasks tackled</i>	<i>Max. time on any subtask (min, s)</i>
1 (I)	2	7,30
1 (II)	4	6,19
2 (I)	5	10,11
2 (II)	3(+2 incorrect)	6,22
3 (I)	3	7,18
3 (II)	2	9,30
Controls	5.7 (± 0.5)	5,35 ($\pm 0,53$)

Results. Two measures of task performance are shown in Table 4. Two different attempts were made by each patient, at least 3 wks apart. On the first attempt Cases 1 and 3 tackled less subtasks than any control and all 3 spent more time on their subtask they attempted for longest (*see* Table 4).

Qualitatively the behaviour of the patients was also atypical. On his first attempt A.P. made notes for over 4 min to help with the dictation task, but in fact never dictated at all. He only attempted 2 subtasks, the second of these occupying the last 7.5 min. On his second attempt he tackled only 4 of the subtasks claiming that he did not dictate his journey back as he had not decided where he would go when he would leave the hospital! Case 2 succeeded in tackling 5 subtasks but spent much longer than any normal subject—over 10 min in 3 separate periods—on 1 of the subtasks (maths I) without ever trying its complementary task (maths II). On his second attempt he behaved bizarrely, changing task 63 times (mean task changes of controls 5.7, SD 2.19) and dictating the two sets of picture names—spending only 14 s and 20 s on them—instead of writing them down; no normal subject carried out a subtask incorrectly. Also on 6 occasions he carried out a subtask immediately after the complementary one, so breaking one of the rules. (One control also made this error. She said she had mistaken the rule as not doing a part I of a pair of subtasks after another part I; this she correctly applied.) Case 3, on her first attempt, only tackled 3 subtasks. On the second she wrote her journey instead of dictating it, attempted only 2 other subtasks and spent the last 9 min on just 1 of them. During this time she looked at the stopwatch on 7 occasions but did not switch tasks.

On the 2 timing tasks Case 1 performed well on both (1.5 min, stopped 1 min 17 s; 2.5 min, stopped 2 min 28 s). Case 2, however, failed the first, stopping at 1 min 38 s although being fully aware he had to stop at 1 min 15 s. On the second task his performance was very poor on both testing occasions. On the first attempt he stopped after 1 min 37 s, saying that he thought 2 min 45 s had elapsed and on his second he failed to stop until 3 min 26 s had elapsed. Case 3 performed well on the 1.25 min task (stopping at 1 min 15 s). However, on the more difficult 2.5 min task she did not stop until 3 min 20 s had passed, despite having looked at the watch at 2 min 38 s. No control was more than 1 s out on the first task or more than 5 s out on the second. (Mean time out for task 1, 0.1 s, SD 0.3; task 2, mean 2.9 s, SD 2.0).

Comment

All 3 patients performed at below the normal range on quantitative measures of performance on the Six Element task. In addition their performance was also qualitatively atypical.

Multiple Errands Test (ME). The purpose of this test, which is undertaken in a pedestrian precinct near the hospital previously unknown to the patients, is for them to carry out a number of tasks in situations where minor unforeseen events can occur. The subtasks the patients had to carry out are basically very simple except for one, which has subcomponents designed to be reasonably demanding for someone of the IQ level and cultural background of the patients. While still inside the hospital the patient is given a card with 8 tasks written on it, 6 of which are simple (e.g., buy a brown loaf, buy a packet of throat pastilles). A seventh requires the subject to be at a certain place 15 min after starting. An eighth is more demanding, 4 sets of information have to be obtained and written on a postcard, namely: (1) the name of the shop in the street likely to have the most expensive item; (2) the price of a pound of tomatoes; (3) the name of the coldest place in Britain yesterday; and (4) the rate of the exchange of the French franc yesterday.

The card also contains the following instructions: 'You are to spend as little money as possible (within reason) and take as little time as possible (without rushing excessively). No shop should be entered other than to buy something. Please tell one or other of us when you leave a shop what you have bought. You are not to use anything not bought on the street (other than a watch) to assist you. You may do the tasks in any order.'

This second part of the instructions—the 'rules'—is then read aloud to the patient who is asked to repeat them. If the rules cannot all be repeated satisfactorily the procedure is repeated until they can be. The subject is then taken the short distance to a shopping precinct where he or she is asked to repeat the rules again. If there is any failure to reproduce the gist of any instruction the subject is reminded of it. It is then indicated to the patient where the limits of the test area are; they are very clearly demarcated by a large street ending in the pedestrian precinct at one end and a set of traffic lights at the other. The behaviour of the subjects while carrying out the activity is monitored by 2 observers and they are also debriefed after the task had been completed.

Controls. Nine control subjects were tested. They were matched with the patients for age (mean 40, range 24–63 yrs) and NART IQ (mean 122, range 113–127).

Results. Table 5 gives the number of errors produced by each of the 3 patients. Each produced at least 2 SDs more than the controls. Table 5 also shows the errors broken down into subcategories: (1) *inefficiencies*—where a more effective strategy could have been applied, e.g., entering the same shop more than once; (2) *rule breaks*—where a specific rule (either social or explicitly mentioned in the task) is broken, e.g., going outside the boundaries or leaving a shop with a newspaper without paying; (3) *interpretation failure*—where the requirements of a particular task are misunderstood, e.g., assuming that the information must be written on the birthday card rather than the postcard; (4) *task failure*—a task either not carried out or not completed satisfactorily.

TABLE 5. ERRORS ON MULTIPLE ERRANDS

	<i>Cases</i>			<i>Control</i>
	<i>1</i>	<i>2</i>	<i>3</i>	
Inefficiencies	6*	9*	5*	1.4 (1.1)
Rule breaks	5*	8*	8*	1.6 (1.3)
Interpretation failures	1	1	1	0.4 (0.7)
Task failures	0	5*	4*	1.1 (1.4)
Total errors	12*	23*	17*	4.6 (2.1)

* More than 2 SD worse than the control subjects.

All 3 patients performed at the 5% level or worse, as estimated from the control data, on both the number of inefficiencies and on the number of rule breaks. There was no difference overall in the rate of errors of the other two types, although Case 3 produced more task failures. Thus 2 normal controls as well as the patients failed to provide the required information on the postcard, probably because the information required was placed on the *back* of the instruction sheet.

Qualitatively the performance of the patients was even worse. They made types of error which were not produced by any control. Case 3 used a clearly irrelevant criterion inappropriately when she broke a rule (entering a shop without buying anything) because when in the shop (a chemist) she found it did not have a soap she especially liked; other cheap soap—which would have been at least as adequate for the task—was available. She also failed to note the time when starting. Cases 1 and 2 both became involved in social complications. Case 1 made the interpretative error of deciding he needed yesterday's newspaper (to find the coldest place in Britain on the previous day). He entered a newsagents and asked if they had the previous day's paper. They had, so he walked out with it, incidentally breaking the buying rule. He was pursued into the street by the shopkeeper who wanted to be paid for it. A.P. had just assumed that because a previous day's newspapers are generally worthless he could have one without paying. Case 2 produced a complex set of errors. Thus one series of actions began with his being the only subject to enter a shop simply to ask about the location of another type of shop (breaking buying rule). He was referred to a shop outside the allowed area, went to that shop (breaking the limit rule) and became involved in

a heated argument with a shop assistant as he asked her to give him a birthday card free (which would have also broken the buying rule)!

Comment

As with the Six Element task, all 3 patients had quantitatively impaired performance, in particular as far as rule breaks and inefficiencies were concerned. Their performance was also qualitatively unusual.

DISCUSSION

We have described 3 patients, each of whom had sustained head injuries in which the area of focal damage lay in frontal regions. The 3 patients performed well on those cognitive tests which were not designed to load specifically on 'frontal functions'. On 'frontal' tests their performance varied. Case 3 performed poorly on a number of such tests. A second (Case 2) performed generally satisfactorily except for a borderline impairment on the Petrides and Milner Self-Ordered Memory. The third (Case 1) performed very satisfactorily on all standard 'frontal' tests. In daily life, however, all 3 appeared unable to act effectively on their own initiative as they organized virtually no nonroutine activities. In addition 2 (Cases 1, 2) were reported to have problems in carrying out multiple subgoal tasks such as shopping and the third (Case 3) avoided such activities.

These difficulties were mirrored in the 2 tests where they had to perform multiple activities, the Six Element Test and Multiple Errands. The 2 tasks required the subject to carry out a number of fairly simple activities in an efficient manner. One, the Six Element, basically involved the devising of a simple plan, scheduling the subtests efficiently and keeping a check on time. The second involved dovetailing the activities in a more real-life setting where minor unforeseen events could occur. All 3 patients performed both quantitatively and qualitatively worse than the normal controls.

If one considers what is involved in carrying out these multiple subgoal tasks, then at a very general level, four basic types of process are relevant. Motivational and memory processes are clearly required and so are a variety of special-purpose cognitive processes of the sort that standard neuropsychological tests assess. In addition there are certain bridge processes which enable the special-purpose cognitive processes to be used to satisfy motivational requirements. A deficit in basic special-purpose cognitive processes seems an implausible explanation of their difficulties on the experimental tasks, given the performance of the patients on the baseline tests. Indeed the most difficult Multiple Errands subtest gave problems for some of the controls as well as the patients; it was the least sensitive part of the procedure. However, frontal patients often manifest inappropriate affect and have frequently been described as apathetic or impulsive (Blumer and Benson, 1975), and also they can have memory problems (e.g., Hécaen, 1964). The possibility of motivational or memory difficulties therefore needs to be considered, especially as the patients, when asked to account for some action, often said that they had completely forgotten their prior intention.

A possible motivational explanation of the impaired performance of the 3 patients is that they require continuous social reinforcement to carry out psychological tasks, and without it their spontaneous motivation would tend to dwindle rapidly; without it they do not persevere. This account is unsatisfactory. Case 1 was given the entire AH6 followed by the Watson-Glaser Critical Thinking Appraisal (Watson and Glaser, 1980)

and left to work through them on his own. He scored at the 50th and 80th percentiles, respectively, the total of the task times being 2 h. Both Cases 2 and 3 carried out the long form of Raven's Matrices working through the test on their own. This took them 31 and 42 min, respectively—less than the 45 min that Raven *et al.* (1977) give as the upper limit of the normal range and yet longer than the 2 multiple-subgoal scheduling tests would normally take. They scored at above the 95th and 80th percentiles, respectively, for their ages on the Matrices.

For the Six Element task a motivational explanation has in any case little immediate face validity. It requires little effort to switch from one simple task to another. More critically, Cases 2 and 3, the subjects for whom a motivational explanation is most plausible on clinical grounds, made errors analogous to the ones made on the Six Element task on an even simpler timing task. When they had to stop carrying out a perceptuomotor task after 2.5 min with no external timing signal but a stopwatch available, they continued after the time limit. Lack of perseverance—the motivational explanation being considered—should presumably not lead to overshoot! Thus motivational factors do not seem likely to be central to their deficit.

Standard retrospective memory processes are also relevant, at least for carrying out the Multiple Errands test where retaining both the layout of the street and the instructions, without using the card, would be helpful. The patients differed in this respect. Case 1 had no difficulties. Case 2 was entirely normal on verbal memory tasks but had visual memory problems which might have affected his performance of the Multiple Errands test; however, they would seem most unlikely to contribute to his difficulty with the Six Element task where there is no spatial memory load and where the instructions are continuously clearly visible on the desk. Case 3 had problems on some memory tasks but these were all in tasks where active organization of the input is required for adequate memory performance and her difficulties appear secondary to her frontal lobe problems (*see* Signoret and Lhermitte, 1976). She was able to carry out memory tasks where the organization was provided for her and in particular she retained text normally, as indicated by her satisfactory performance on Logical Memory where the amount she retained showed no decline over an hour's delay. Like the other patients, she was able to recite correctly the instructions for the tasks in her own words after they had been presented to her. This makes it most unlikely that retrospective memory difficulties lie at the root of her problems, at least on the Six Element test.

Given that the problems these 3 patients exhibit on these multiple subgoal tests do not fit well with an assumption of basic motivational impairments on the one hand or of special purpose cognitive or retrospective memory processes on the other, do they fit with impairments to the bridge processes that lie between the two? To assess this it is necessary to consider the bridge processes in more detail.

These processes lie in the domain of the creation and maintenance of goals and intentions, of their realization at appropriate times (prospective memory (Meacham and Leiman, 1982; Kvavilashvili, 1987)) and of planning. This is an area where well developed theories are lacking. However, a position developed by one of us in collaboration with D. A. Norman is that where a task cannot be adequately carried out through the application of well-learned action or thought routines alone, it requires the use of a Supervisory System, which is anteriorly located in the cortex (Norman and Shallice, 1980, 1986; *see also* Shallice, 1988). We will provisionally follow this approach

and assume in line with it that the performance of a basic routine activity, like paying the assistant when shopping, is not dependent on anterior systems once it has been selected for operation. What processes would a Supervisory System control?

It is often argued that when routine actions will not achieve a goal satisfactorily the problem-solving process required involves at least four stages and that these may be called in a recursive fashion (de Groot, 1965; Luria, 1966; Ben-Yishay and Diller, 1983). One or more potentially achievable goals must be specified: a course of action (a plan) must be drawn up, a solution attempt is then executed and finally the provisional solution assessed.

It is now argued that in normal human activity plans when originally developed do not correspond to completely worked out courses of action (Suchman, 1987; Ellis, 1989). A consequence is that planning as a whole needs to be viewed as a more opportunistic process carried out on-line when opportunities arise or difficulties occur. There must, however, be processes related to intentions, which link the schematic plan with the potential to realize it effectively at some later time. We propose that these processes operate through the activating of markers and their later triggering to interrupt ongoing behaviour at a future time. A marker is basically a message that some future behaviour or event should not be treated as routine and instead, some particular aspect of the situation should be viewed as especially relevant for action. If the behaviour or event does occur later, then the marker would be triggered and this would lead to inhibition of the activity being carried out, the reassessment of the situation and so potentially the switching in or out of a particular course of action linked to the marker. Thus a task rule, such as not entering a shop unless something is to be bought, would be realized through a marker becoming activated when the instructions are understood, that when about to enter a shop a check should be made that something is to be bought. This marker would then be triggered by the action of approaching a shop or thinking of doing so. Many markers could be simultaneously activated.

On this view the following processes are plausibly controlled by a Supervisory System, although all are also likely to involve other systems (1) goal articulation; (2) provisional plan formulation; (3) marker creation; (4) marker triggering (by an appropriate mental or physical event). In addition, an evaluation process can occur at any stage and if the result is that the present situation or solution is unsatisfactory this would lead to articulation

TABLE 6. SIMPLE SCENARIO TO ILLUSTRATE COMPONENT PROCESSES IN PRACTICAL PROBLEM SOLUTION

7.45 a.m.	Situation: receive urgent letter.
7.50 a.m.	Goal Articulation: to reply as soon as possible (NB, no stamp available).
8.00 a.m.	Plan formulation: decide to buy stamps at Post Office on way to work and write letter immediately on arrival and post it at lunchtime. Marker Creation 1—When leaving underground — Go to P.O. Marker Creation 2—When arrive work — Write brief letter (if not possible on arrival—carry out Plan Modification). Marker Creation 3—When go to lunch — Post letter.
9.00 a.m.	(Leave Underground) Marker Trigger 1 — Go to P.O.
9.15 a.m.	(Arrival work) Marker Trigger 2 — Write letter*.
12.30 a.m.	(Go lunch) Marker Trigger 3 — Post letter.

* If not possible to write letter at that time plan modification occurs and new markers set up.

of unforeseen subgoals and hence plan modification; this would also be necessary if unforeseen events occur. These stages are illustrated in the simple scenario shown in Table 6.

The present 2 multiple-subgoal tests appear to require all these processes and many of the errors that occur in the execution of the tests appear to occur from a failure of one or other of them, although quite frequently it is not possible to be precise about which process is impaired. However, if one groups the above processes into three types then a provisional grouping of the lapses that occur in the 2 experimental tasks can be made, although possible overlaps between the three types still exist. The types are: (1) Plan Formulation or Modification; (2) Marker Creation or Triggering; (3) Evaluation and Goal Articulation (*see* Table 6 for an example). Table 7 gives examples from the

TABLE 7. EXAMPLE OF THE THREE TYPES OF LAPSE MADE BY THE PATIENTS ON THE EXPERIMENTAL TASKS

<i>Plan formulation or modification</i>	<i>Marker creation or triggering</i>
1. Going to Post Office before all relevant information obtained (Cases 1–3)	1. A very extended period spent on a single SE subtask without a break (Cases 1, 3)
2. No plan for SE (Cases 1–3)	2. Going out of bounds in ME (Cases 2, 3)
3. Failure to look at watch at start of ME (Case 3)	3. Attempt not to pay for item in shop (Case 1, 2)
4. Not knowing quite how he would get to X and therefore not tackling SE. Journey back subtask (Case 1)	4. Looking in irrelevant shops (Cases 1, 2)
5. Generation of inappropriate criteria in an ME subtask (Case 3)	5. Not using the notes made for 4 min about journey at start of SE
<i>Evaluation and goal articulation</i>	
1. Posting postcard without all information (Cases 2, 3)	
2. Failure to keep a check of money in ME (Cases 2, 3)	
3. Thinking finished ME when had not (Case 1)	
4. Ending SE attempt 1 min early when far from having completed subtasks (Case 3)	
5. Carrying out SE subtasks incorrectly (Cases 2, 3)	

SE = Six Elements; ME = Multiple Errands.

2 experimental tasks of the three types of lapse. It can be seen that there are examples of all three types being made by each of the patients, but the prevalence of rule breaks in the behaviour of all 3 patients suggest that marker errors are a particularly important problem. It is also apparent that the marker errors can be subdivided. In some (1, 5) the marker has to respond to a temporal cue, in some to a social convention cue (3) and in some to a task rule cue (2, 4). In all these cases triggering the marker should lead to the inhibition of the ongoing behaviour, but in other cases triggering the marker should have led to a positive behaviour. However, for this type of error there was much more overlap with control subjects (e.g., not coming to tell E when something was bought in a shop (all 3 patients, but also 5/9 controls)).

If the everyday life behaviour of the patients is examined, it is often less easy to categorize many of the lapses that they make because they involve a larger period of time. Nevertheless, the problems they manifest seem likely to arise from the same source as their difficulties in the experimental task.

The processes of plan formulation and evaluation have often been ascribed to frontal

lobe systems (Penfield and Evans, 1935; Luria, 1966) as has goal articulation (Duncan, 1986) and together with marker creation and triggering, these processes are key elements in the temporal integration of behaviour which Fuster (1980) argued to be the basic function of the frontal lobes. To our knowledge, however, no experimental setting has been investigated in which errors occurred that could be clearly ascribed to impairments in these processes in patients whose other cognitive functions were shown to be essentially intact. If the errors of marker creation or triggering are considered, the failure to interrupt ongoing behaviour and produce a specifically relevant action is related to the difficulty of inhibiting central sets ascribed to frontal systems by Rosvold and Mishkin (1961), but the effects of marker creation and triggering need not all be inhibitory in nature.

Overall, plan formulation/modification, marker creation and triggering and evaluation and goal articulation may be considered complementary component processes undertaken by the Supervisory System on the model of Norman and Shallice (1980), and it is these processes that are impaired in the Strategy Application Disorder. Whether it is possible for the individual components to be selectively impaired is not possible to assess with the current evidence. However, in Case 1, marker creation and triggering processes seemed the most affected, while in Case 3 it was goal articulation, plan formulation and evaluation. It therefore seems a plausible possibility that finer fractionation of component processes undertaken by the Supervisory System can occur.

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APPENDIX. SIX ELEMENT TASK INSTRUCTIONS

In the next 15 min we would like you to carry out 3 different tasks, each of which is in 2 parts. The tasks are:

1. To dictate* into the tape-recorder a *brief* account of two journeys. (a) Your journey to here; (b) your intended journey from here.
2. To write down the names of as many pictures as you can (in order). (a) Those in the left pile (set A); (b) those in the right pile (set B).
3. To solve 2 sets of arithmetic problems (as many as you can, in order). (a) Set I; (b) set II.

But you are not allowed to do the 2 subtasks (a) and (b) of the same type one after the other. Each of the 6 subtasks is given equal weight. Within subtasks 2 and 3, points will be given for correct answers. Earlier pictures/problems will be given more points than later ones in each subtask *and* errors/omissions will be penalized.

* The cassette recorder records throughout the 15 min so that the subject does not need to operate it.